CHAPTER 4

Ensemble of Uncoupled Oscillators

- The optical properties of semiconductors are determined by the interaction of various types of oscillators and the electromagnetic field
- An incident electromagnetic field will cause oscillators to perform forced oscillations – depend on
  - the frequency of the incident light
  - the eigenfrequency of the oscillators
  - the coupling strength
  - the damping of the oscillation

Main content

- 4.1 Equations of Motion and the Dielectric Function Based on Classic Theory
- 4.2 Corrections Due to Quantum Mechanics and Local Fields
- 4.3 Spectra of the Dielectric Function and of the Complex Index of Refraction
- 4.4 The Spectra of Reflection and Transmission

4.1 Equations of Motion and the Dielectric Function

- Assume that: An ensemble of identical uncoupled harmonic oscillators, neglecting damping for the moment.
- Then eigen-frequency ω_0':

$$\omega_0'^2 = \beta m^{-1} \tag{4.1}$$

- Fig.4.1b: $\lambda = \infty \quad or \quad k = 0$ (4.2a)
- Fig.4.1c: $\lambda_{\min} = 2a \quad or \quad k_{\max} = \frac{\pi}{a}$ (4.2b)

1. Γ-point： $k = 0$ gives the center of the first Brillouin zone
2. (4.2b): the shortest physically meaningful wavelength and the largest k -vector gives the boundary of the first Brillouin zone for a simple linear chain or cubic lattice.

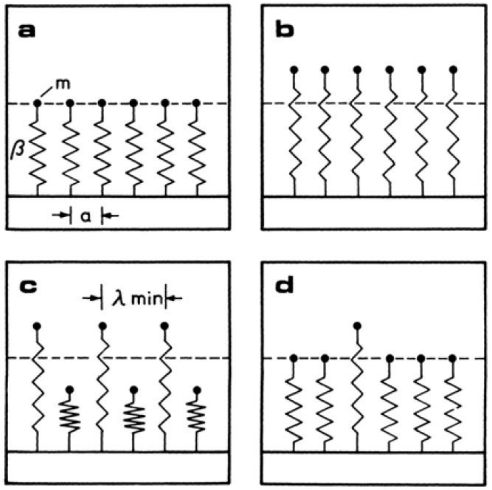


Fig. 4.1. A part of a periodic array of uncoupled oscillators in their equilibrium position (**a**); elongated with a wavelength $\lambda \Rightarrow \infty$ (**b**); with the shortest physically meaningful wavelength $\lambda_{\min}$ (**c**), and a non-propagating wave packet (**d**). After [63H1]

Uncoupled and coupled oscillators

$$\begin{aligned} &zero \quad coupling \leftrightarrow zero \quad bandwidthor \leftrightarrow \upsilon_g = 0, Fig.4.2a \\ &finite \quad coupling \leftrightarrow finite \quad bandwidthor \leftrightarrow \upsilon_g \neq 0, Fig.4.2b \end{aligned} \tag{4.3}$$

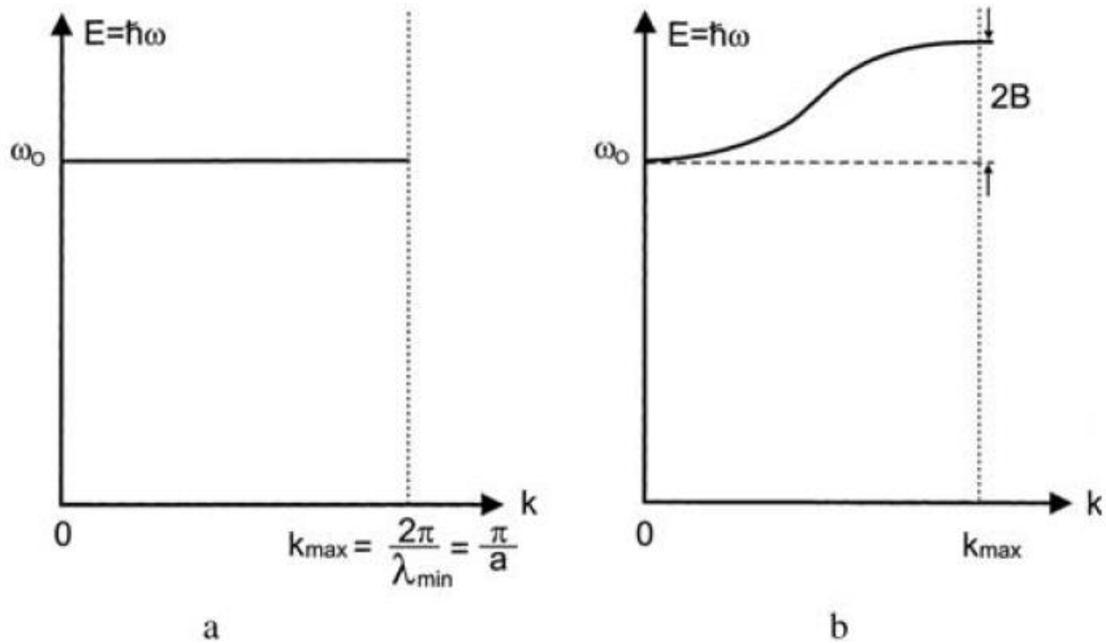


Fig. 4.2. The dispersion relation of an ensemble of uncoupled (**a**) and of coupled (**b**) oscillators

- The electric field of incident light on the independent oscillators:

$$\boldsymbol{E} = \left(E_0, 0, 0\right)\exp\left[i\left(k_z z - \omega t\right)\right] \quad (4.4a)$$

Considering the oscillator at $z = 0$:

$$\boldsymbol{E} = \left(E_0, 0, 0\right)e^{-i\omega t} \quad (4.4b)$$

Assume every oscillator carries a charge e, then a force by $\boldsymbol{E}$:

$$F = eE_0 e^{-i\omega t} \quad (4.5)$$

The equation of motion is:

$$m\ddot{x} + \gamma m\dot{x} + \beta x = eE_0 e^{-i\omega t} \quad (4.6)$$

The general solution of (4.6) =the general solution of the corresponding homogeneous equation + a special solution of the inhomogeneous one.

Assume :
$$x(t) = \boxed{x_0 \exp\left[-i\left(\omega_0'^2 - \gamma^2/4\right)^{1/2} t\right]\exp\left(-t\gamma/2\right)} + \boxed{x_p e^{-i\omega t}} \quad (4.7)$$

For $\omega_0'^2 - \gamma^2/4 > 0$: defines the regime of weak damping.

The solution of the homogeneous equation describes a transient feature

The second part in Eq (4.7) is a forced oscillation, substitute it into (4.6):

$$x_p = \frac{eE_0}{m}\left(\omega_0'^2 - \omega^2 - i\omega\gamma\right)^{-1} \quad (4.9)$$

The dipole moment :

$$p_x = ex_p \quad (4.10)$$

- The polarizability: $\hat{a}(\omega)=\frac{ex_p}{E_0}=\frac{e^2}{m}\left(\omega_0'^2-\omega^2-i\omega\gamma\right)^{-1}$ (4.11)
- The polarization density ***P*** with amplitude ***P***$_0$:

$$\boldsymbol{P}_0 = N\hat{a}\mathbf{E}_0 = \frac{Ne^2}{m}\left(\omega'^2{}_0-\omega^2-i\omega\gamma\right)^{-1}E_0 \qquad (4.12)$$

- This means that we describe the ensemble of oscillators as an effective medium, an approach which is well justified for a«λ
- From (4.12) and (2.27): we get for the dielectric displacement:

$$\boldsymbol{D}=\varepsilon_0\boldsymbol{E}+\boldsymbol{P}=\varepsilon_0\left[1+\frac{Ne^2}{\varepsilon_0 m}\left(\omega_0'^2-\omega^2-i\omega\gamma\right)^{-1}\right]\boldsymbol{E} \qquad (4.13)$$

$$\varepsilon(\omega)=1+\frac{Ne^2}{\varepsilon_0 m}\left(\omega_0'^2-\omega^2-i\omega\gamma\right)^{-1}=\chi(\omega)+1 \qquad (4.14)$$

4.2 Corrections Due to Quantum Mechanics and Local Fields

- In classic mechanics model, the term Ne^2m^{-1} gives the coupling strength
- In quantum mechanics, this coupling is given by the square of transition matrix element for dipole-allowed transitions :

$$\left|H_{ij}^{D}\right|^2 = \left|\left\langle j\left|H^{D}\right|i\right\rangle\right|^2 \quad (4.15)$$

- Usually define the oscillator strength : $f = \frac{2N\omega_0'}{\varepsilon_0\hbar}\left|H_{ij}^{D}\right|^2$ (4.16)

  and

$$\varepsilon(\omega) = 1 + \frac{f}{\omega_0'^2 - \omega^2 - i\omega\gamma} \quad (4.17)$$

- For dilute systems, *E* is just the external incident field and use (4.17) as it is
- For dense systems, consists of two parts = external field and the field created by all the other dipoles. $\varepsilon(\omega)$ can be written :

$$\varepsilon(\omega) = 1 + \frac{f}{\omega^2{}_0 - \omega^2 - i\omega\gamma} \quad (4.19)$$

with a shifted eigenfrequency :

$$\omega^2{}_0 = \omega'^2{}_0 - \frac{Ne^2}{3m\varepsilon_0} = \omega'^2{}_0 - \frac{f}{3} \quad (4.18b)$$

4.3 Spectra of the Dielectric Function and of the Complex Index of Refraction

- A semiconductor contains not only one type of oscillators
- Helmholtz-Ketteler formula or Kramers-Heisenberg dielectric function:

$$\varepsilon(\omega) = 1 + \sum_j \frac{f_j}{{\omega^2}_{0j} - \omega^2 - i\omega\gamma_j} \tag{4.20a}$$

If the eigenfrequencies ω_0 form a continuous band:

$$\varepsilon(\omega) = 1 + \int \frac{f(\omega_0)}{\omega_0^2 - \omega^2 - i\omega\gamma(\omega_0)} d\omega_0 \tag{4.20b}$$

For simplicity, we use (4.20a) in the following.

- For a single resonance ω_{0j} well separated from all other resonances, the contribution of other resonance $\omega_{0j'}$ in the vicinity of ω_{0j}

For $\omega << \omega_{0j'}$, $\frac{f_j}{\omega_{0j'}^2} = const$ (4.21)

and for the contribution tends to zero for $\omega >> \omega_{0j'}$

So we can neglect the lower resonance and summarize all higher resonances in a so-called background dielectric constant ε_b

So we get the simplified expression in the surroundings of ω_{0j} :

$$\varepsilon(\omega) = \varepsilon_b + \frac{f_j}{{\omega^2}_{0j} - \omega^2 - i\omega\gamma_j} \tag{4.22a}$$

- Separated into real and imaginary parts:

$$\varepsilon(\omega)=\left(\varepsilon_b+\frac{f\left(\omega_0^2-\omega^2\right)}{\left(\omega_0^2-\omega^2\right)^2+\omega^2\gamma^2}+i\frac{\omega\gamma f}{\left(\omega_0^2-\omega^2\right)^2+\omega^2\gamma^2}\right) \qquad (4.22b)$$

$$=\varepsilon_1(\omega)+i\varepsilon_2(\omega)$$

For $\gamma\Rightarrow 0$, $\left(\omega_0^2-\omega^2\right)=\left(\omega_0+\omega\right)\left(\omega_0-\omega\right)\approx 2\omega_0\left(\omega_0-\omega\right)$, we get :

$$\varepsilon(\omega)=\varepsilon_b+\frac{f}{2\omega_0}\frac{1}{\omega_0-\omega} \qquad (4.22c)$$

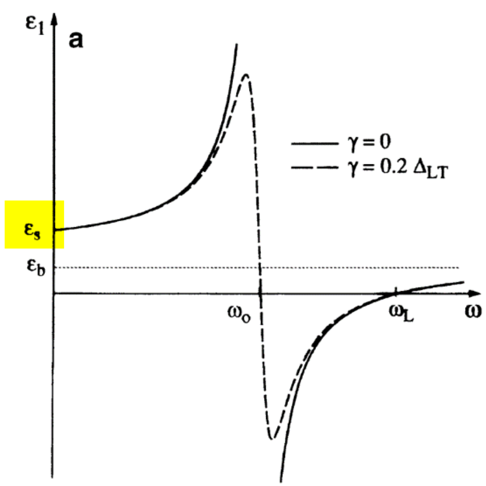
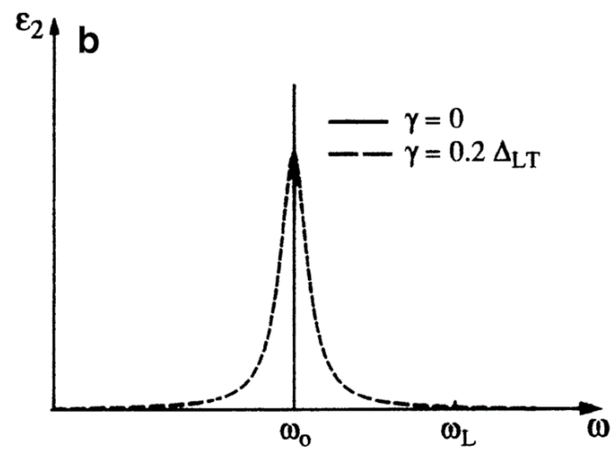


Fig. 4.3 The real (**a**) and imaginary (**b**) parts of the dielectric function for zero and finite damping



$\gamma\rightarrow 0$, there is a pole in $\mathrm{Re}\{\varepsilon(\omega)\}$, and $\mathrm{Im}\{\varepsilon(\omega)\}$ converges to a δ function at ω_0
Finite damping results in a broadening of $\mathrm{Im}\{\varepsilon(\omega)\}$ to the Lorentzian lineshape of (4.22b) and a smooth connection of the two branches of $\mathrm{Re}\{\varepsilon(\omega)\}$

- Small damping :

  For $\gamma \Rightarrow 0$, we find: $\varepsilon(\omega = \omega_T = \omega_0) = \infty$ connected with the singularity.

  $\varepsilon(\omega = \omega_L) = 0$ The solution of $\nabla \cdot \boldsymbol{D} = \varepsilon_0 \nabla \cdot \varepsilon(\omega) \cdot \boldsymbol{E} = 0$ (4.23)

- The relation between these two frequencies:

  $$\omega_L^2 - \omega_T^2 = f / \varepsilon_b \sim \left|H_{ij}^D\right|^2 \quad (4.25)$$

  So : $\left|H_{ij}^D\right|^2 \neq 0 \Leftrightarrow f \neq 0 \Leftrightarrow \Delta_{LT} = \hbar(\omega_L - \omega_T) \neq 0$ (4.26)

  See Fig.4.4 : a longitudinal electric field, acts as an additional restoring force and increases the longitudinal eigenfrequency ω_L above the transverse eigenfrequency ω_T

- The static dielectric constant ε_s below ω_0 and the value above ε_b: $\varepsilon_s = \varepsilon_b + \frac{f}{\omega_0^2}$ (4.27)

  and the Lyddane-Sachs-Teller relation:

  $$\frac{\varepsilon_s}{\varepsilon_b} = \frac{\omega_L^2}{\omega_0^2} > 1 \quad for \quad f > 0 \quad (4.28)$$

- The term "small damping":

  $$\gamma < \hbar^{-1}\Delta_{LT} = \omega_L - \omega_T \quad (4.29)$$

  and frequency ω far above or below ω_0 means:

  $$\hbar\left|\omega - \omega_0\right| >> \Delta_{LT} \quad (4.30)$$

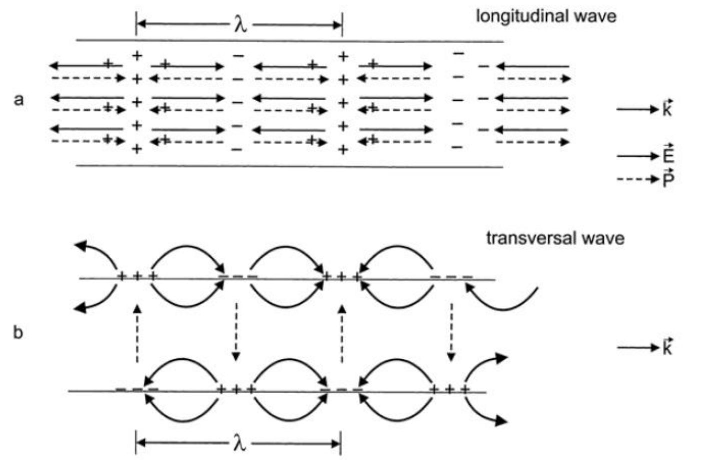


Fig. 4.4. The polarisation charges and the resulting electric and polarisation fields (*solid* and *dashed arrows*, respectively) in a slab of matter for a longitudinal (a) and a transversal wave (b)

- The complex index of refraction : $\tilde{n}(\omega)=n(\omega)+i\kappa(\omega)$ $\quad \tilde{n}(\omega)=\varepsilon^{1/2}(\omega)$

consequently: $\varepsilon_1(\omega)=n^2(\omega)-\kappa^2(\omega)$ $\quad \varepsilon_2(\omega)=2n(\omega)\kappa(\omega)$

or : $n(\omega)=\left(\frac{1}{2}\left\{\varepsilon_1(\omega)+\left[\varepsilon_1^2(\omega)+\varepsilon_2^2(\omega)\right]^{1/2}\right\}\right)^{1/2}$ $\quad \kappa(\omega)=\left(\frac{1}{2}\left\{-\varepsilon_1(\omega)+\left[\varepsilon_1^2(\omega)+\varepsilon_2^2(\omega)\right]^{1/2}\right\}\right)^{1/2}$ (4.32)

When light frequency approaching the resonance from the low-frequency side, *n* increase drastically. Between ω_0 and ω_L we find

$$\omega_0 \le \omega \le \omega_L \begin{cases} n = 0 & \text{for} \quad \gamma = 0 \\ n \ll 1 & \text{for} \quad \gamma \ne 0 \\ \kappa > n \end{cases} \tag{4.33}$$

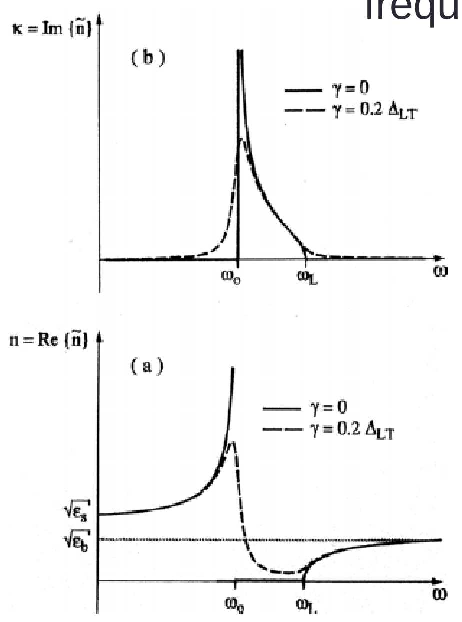


Fig. 4.5. Real (**a**) and imaginary (**b**) parts of the complex index of refraction for zero and finite damping

For $\gamma=0$: there is no propagating or wave-like solution, only a spatially exponentially decaying amplitude similar to the type known for total reflection in the medium with the lower index of refraction

For finite γ : some light can penetrate into the medium, but the light is damped over a distance shorter than the wavelength.

4.4 The Spectra of Reflection and Transmission

- Now discuss the spectra of reflection and transmission:
- Assume a single interface between vacuum(or air) and the medium, and normal incidence.

$$R(\omega)=\frac{I_r}{I_i}=\frac{\left[n(\omega)-1\right]^2+\kappa^2(\omega)}{\left[n(\omega)+1\right]^2+\kappa^2(\omega)}$$

Stop-band

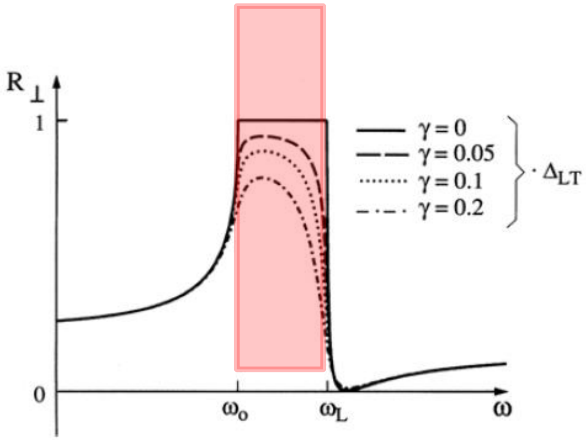


Fig. 4.6. The reflection spectrum of a single resonance with zero and finite damping for normal incidence

$$R(\omega)=\frac{\left(\varepsilon_s^{1/2}-1\right)^2}{\left(\varepsilon_s^{1/2}+1\right)^2} \quad for \quad \omega_0-\omega>>\frac{1}{\hbar}\Delta_{LT}$$

R starts with an almost a constant value

$$R(\omega)=\frac{\left(\varepsilon_b^{1/2}-1\right)^2}{\left(\varepsilon_b^{1/2}+1\right)^2} \quad for \quad \omega-\omega_0>>\frac{1}{\hbar}\Delta_{LT}$$

R converges to a lower constant value

- For vanishing damping, R increases just below ω_0 and reaches the value R=1 for $\omega=\omega_0$. Between ω_0 and ω_L R=1. For $\omega>\omega_L$ R drops rapidly and reaches 0 at the frequency wheren *n*=1.Then R increases again towards a constant value
- For finite damping the R(ω) is smoothed out due to the finite values of n for $\omega_0\leq\omega\leq\omega_L$. The reflection minimum occurs also for weak damping only slightly above ω_L, but the reflection maximum is no longer related to ω_0

Reststrahlenbande

- The spectral region between ω_0 and ω_L is often called "Reststrahlenbande" for the following reason. If we send a light beam with a broad, essentially flat, spectral distribution of the intensity around ω_0 onto a sample and allow it to be reflected several times, e.g., in the configuration given in Fig. 4.7a, then significant intensity will remain as a "rest" only in the region

$$\omega_0 \leq \omega \leq \omega_L$$

- It should be noted that we do not assume any coherent superposition of the beams, i.e., the Reststrahlenbande comes only from $\varepsilon(\omega)$ and has nothing to do with Fabry–Perot resonators

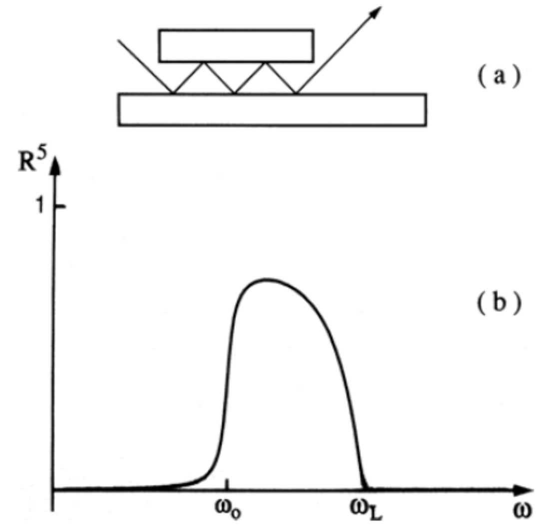


Fig. 4.7. An arrangement for multiple reflection from a medium (a) and the resulting "Reststrahlbande" (b)

α_c

Semiconductor slab

- Handle semiconductor slabs with two surfaces and a geometrical thickness d
- (a) No interference
- (b) Interference

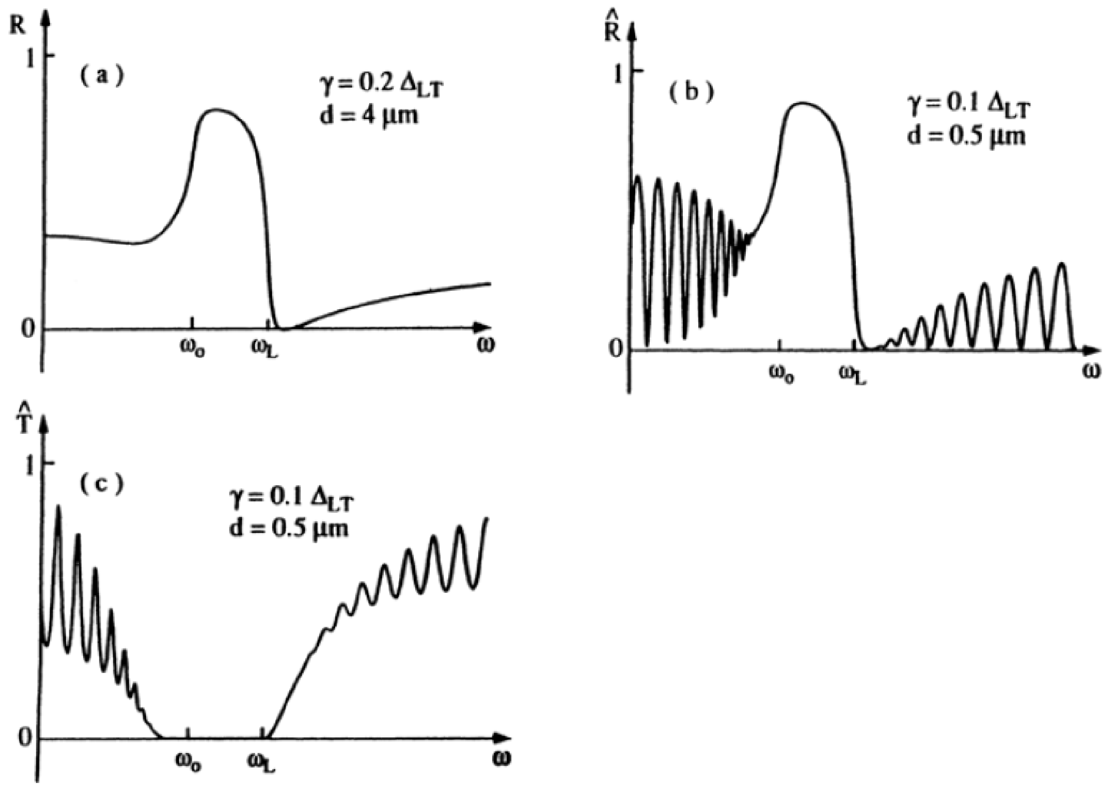


Fig. 4.8. The reflectivity $\hat{R}$ of a thin slab of matter in the vicinity of a resonance for the cases where intensities (**a**) or field amplitudes (**b**) have to be added and the transmission (**b**) for the case where amplitudes add

4.5 Interaction of Close Lying Resonances

- Discuss what happens for two close-lying resonances.
  For instance :for the A and B Γ_5 excitons in ZnO.
- We consider two resonances with equal oscillator strength f and slightly $\omega_A < \omega_B$.
  and we neglect damping and spatial dispersion.

$$\varepsilon_A(\hbar\omega) = \varepsilon_b + \frac{f}{\hbar^2(\omega_A^2 - \omega^2)}, \quad (4.37a)$$

$$\varepsilon_B(\hbar\omega) = \varepsilon_b + \frac{f}{\hbar^2(\omega_B^2 - \omega^2)}, \text{and} \quad (4.38a)$$

$$\varepsilon_{tot}(\hbar\omega) = \varepsilon_b + \frac{f}{\hbar^2(\omega_A^2 - \omega^2)} + \frac{f}{\hbar^2(\omega_B^2 - \omega^2)} \quad (4.38b)$$

$$= \varepsilon_A(\hbar\omega) + \varepsilon_B(\hbar\omega) - \varepsilon_b \quad (4.38c)$$

The situation becomes significantly different for (4.38c) shown in Fig.4.9c.

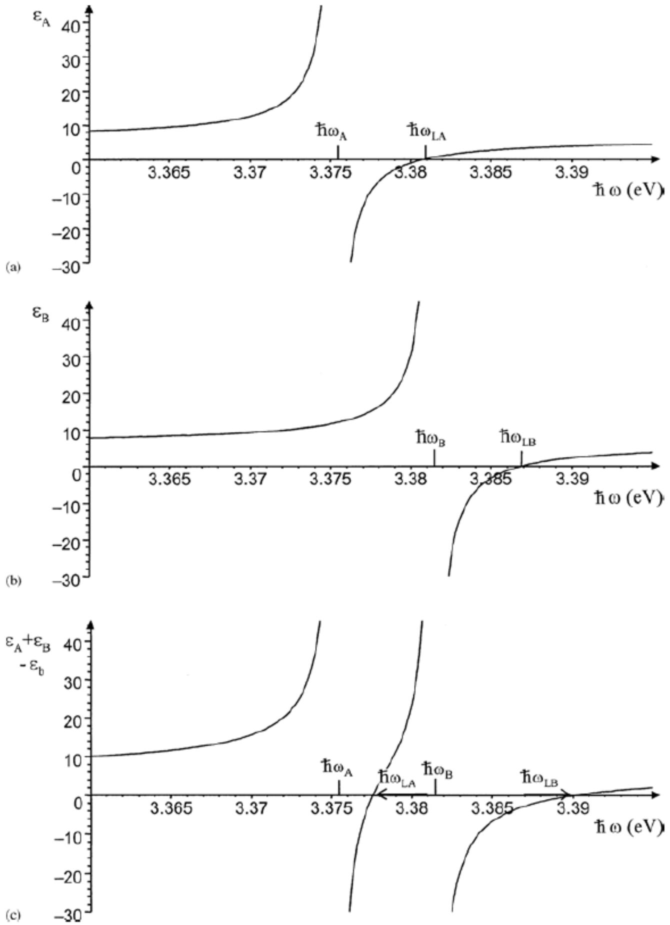


Fig. 4.9. The real part of $\varepsilon(\hbar\omega)$ for two resonances separately (a, b) and for two close lying resonances (c). Numbers on the x-axis refer roughly to ZnO